DIGITAL WORKFORCE GOVERNANCE | FACE RECOGNITION ACCESS

Contract Labour Management Success Story

100% customized software. Verified people. Compliant access.



FROM DOCUMENTS TO GATE DECISIONS - ONE DIGITAL CONTROL LAYER

5,000+	4 + 1	USD 0.048M	3 YEARS
ACTIVE WORKER CAPACITY	GATE DEVICES + SPARE	APPROX. PROJECT VALUE	WARRANTY + CAMC

CLIENT: PETRONET LNG LIMITED | DAHEJ LNG TERMINAL | DESIGN, DEVELOPMENT, DELIVERY & SUPPORT BY D-TECH

Conceptual access-process image shown for communication. Installed-site evidence is included inside.

THE TRANSFORMATION

From manual passes to governed digital access

D-TECH designed, developed and delivered a standalone face-recognition access platform that connects contractor compliance, worker identity, gate-pass validity, turnstile decisions and workforce reporting.

5,000+	1,000	2	5 YR
ACTIVE PASSES	REUSABLE RFID CARDS	REDUNDANT SERVERS	DATA BACKUP

THE CORE CHALLENGE

Gate access had to be both fast and defensible.

Every contract worker needed a unique identity, a valid work request, current safety and medical records, police verification, contractor-linked labour-licence capacity and area-specific authorization. Security also required real-time entry/exit control, traceable exceptions and auditable reports.

01 One identity

Face biometric helps stop duplicate or renamed worker registrations.

02 One rule engine

Pass validity follows the shortest valid compliance credential.

03 One access loop

Entry and exit events remain sequenced and accountable.

04 One evidence trail

Searchable events, passes, hours and exceptions support audits.

Result

A scalable, industrial-grade digital labour-governance system that improves security, compliance visibility and gate productivity without depending on the corporate WAN/LAN.

Business Requirements

CONTROL OBJECTIVES

The requirement was governance - not only attendance

The customer needed every gate decision to be driven by an approved worker profile, active documents and a valid area-specific pass.

01 Contractor governance

Control active passes against each contractor's labour-licence capacity and work order.

02 Identity assurance

Detect duplicate face identities, renamed records and cross-contractor re-registration attempts.

03 Compliance validity

Link pass duration to requisition, medical fitness, safety training and police verification.

04 Area authorization

Differentiate Non-Ops, Operations, Jetty and All-Area permissions with visible pass coding.

05 Gate discipline

Prevent invalid access and preserve a complete entry-exit loop through the turnstiles.

06 Operational resilience

Run standalone with redundant servers, power backup, local infrastructure and spare device.

07 Audit-ready reporting

Provide manpower, work-hour, overtime, pass, entry-exit and exception reports.

08 Lifecycle support

Include training, warranty, remote/on-site support, CAMC and long-term spares availability.

DESIGN PRINCIPLE

No compliant identity, no valid pass. No valid pass, no authorized turnstile command.

End-to-End Digital Workflow

FROM REQUEST TO AUDIT

A controlled seven-stage worker lifecycle

The workflow brings contractor administration, statutory compliance, identity verification, physical access and reporting into one governed process.

01	**Contractor**	Work order and labour-licence master
02	**Worker**	Profile, ID and area request
03	**Compliance**	Police, medical and safety checks
04	**Approval**	Validity and area authorization
05	**Enrolment**	Face template and optional RFID
06	**Gate**	Live face match and turnstile decision
07	**Evidence**	Hours, exceptions and audit reports

AUTOMATED DECISION

Identity match + valid pass + permitted area = access granted

If any rule fails, the system denies the transaction, records the event and presents an operator-visible exception for controlled action.

CONTROLLED MODES Face only | RFID + Face | RFID only | Admin-controlled exception

Digital Worker Enrolment

100% CUSTOMIZED SOFTWARE

One worker profile. Every compliance check.

The enrolment desk converts approved documents and worker identity into a governed digital pass record.



OK **Government ID**
Identity master

OK **Labour licence**
Contractor capacity

OK **Police verification**
Validity rule

OK **Medical fitness**
Validity rule

OK **Safety training**
Validity rule

OK **Work order**
Contract link

OUTPUT **Unique worker record + face template + compliance validity + area permissions + photo gate pass + complete approval history.**

Face-Based Gate Access

REAL-TIME CONTROL

A live identity decision at every entry and exit

The worker presents at the protected face terminal. The software validates identity, pass status, area permission and gate sequence before commanding the turnstile.



1
LIVE FACE
Physical person detected and matched

2
PASS STATUS
Active validity and contractor link

3
AREA RULE
Gate and zone authorization

4
TURNSTILE
Open or deny with logged reason

AUTHORIZED

Turnstile command issued. Time-in/time-out event stored. Dashboard count updated.

DENIED / EXCEPTION

Expired, blocked, duplicate, missing entry, invalid area or unrecognized face is logged for action.

Customized Software Platform

APPLICATION CAPABILITY

A complete contractor-worker control suite

The software is purpose-built around plant security, statutory validity, pass administration, access control and workforce analytics.

01 Contractor Master

Work order, licence limits and contractor-linked active passes.

02 Worker Master

Profile, identity, address, blood group, photo and permissions.

03 Compliance Engine

Police, medical, safety and requisition validity calculations.

04 Pass Management

Unique number, colour-coded area, print, extend, block and unblock.

05 Biometric Identity

Face enrolment, duplicate-person alert and optional RFID binding.

06 Gate Control

Entry/exit sequence, live pass validation and turnstile command.

07 Exception Console

Admin-controlled corrections for defined technical contingencies.

08 Reports & Exports

PDF/Excel/Word event, hours, pass, contractor and audit outputs.

09 Backup & Archive

Three-year live records with five-year backup and retrieval.

STANDALONE SECURITY ARCHITECTURE

Dedicated software and hardware operate independently from the corporate security WAN/LAN, supporting controlled local administration.

AUTOMATED VALIDITY

The pass expires when the earliest requirement expires

The rule engine protects the gate from relying on a visually valid card whose underlying approval, medical fitness or safety credential has already expired.

CONTROL	VALIDITY / RULE	GATE IMPACT
Pass requisition	Requested duration	Controls the approved work period
Police verification	Local / hometown rule	Controls longer-duration access
Medical fitness	Six-month validity	Worker health credential
Safety training	Six-month validity	Plant safety authorization
Labour licence	Contractor capacity	Blocks passes above approved limit
Work order	Contractor / PO mapping	Links access to authorized work
Area permission	Non-Ops / Ops / Jetty / All	Restricts worker to authorized zones
Entry-exit sequence	Complete punch loop	Prevents inconsistent movement records

VALIDITY FORMULA

Gate-pass validity = earliest expiry among all mandatory controls

This makes compliance operational at the gate instead of leaving it only in files and spreadsheets.

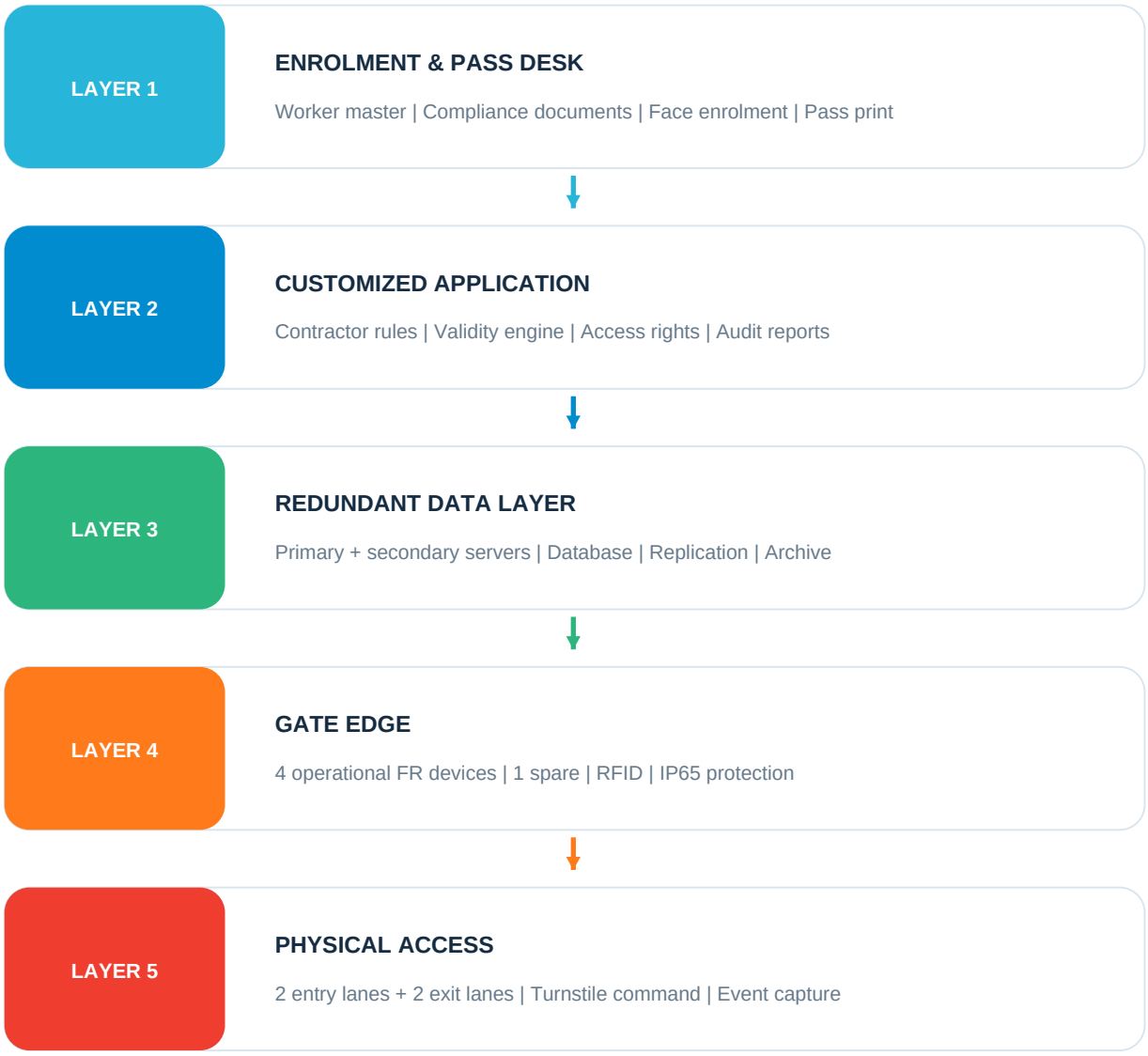
CUSTOMER HELP **Fewer expired credentials, fewer unauthorized entries, faster audits and clearer accountability.**

Delivered Solution Architecture

TURNKEY DEPLOYMENT

A resilient standalone platform built for industrial gates

D-TECH delivered the application, biometric devices, protected field installation, redundant compute, access integration, networking, power backup, training and lifecycle support.



RESILIENCE BUILT IN

Local standalone operation | dual-server replication | UPS backup | spare FR device | five-year spares support | remote and on-site service.

Installed-Site Evidence

FIELD DELIVERY

Face recognition protected for real gate conditions

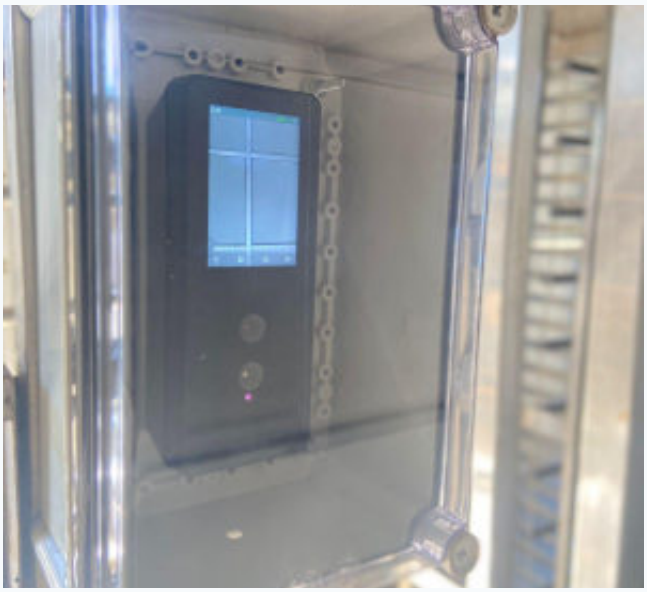
Installed devices are mounted beside the full-height turnstiles in protective enclosures, with field cabling and access-control integration completed at the operating gate.



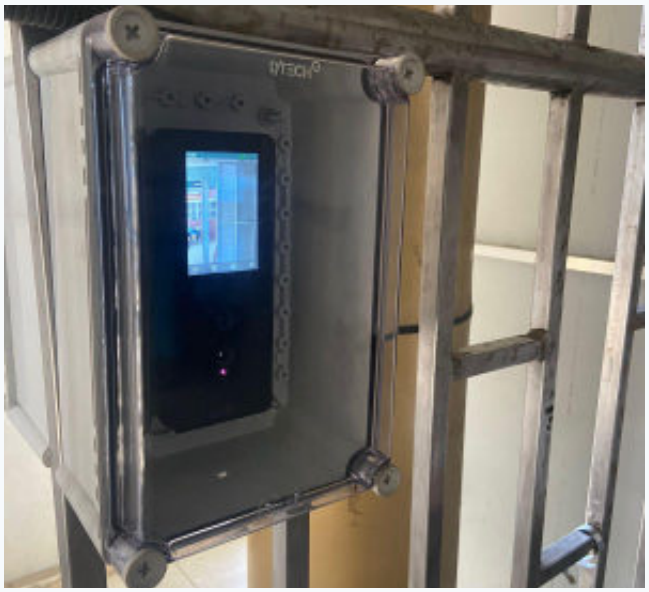
FR terminal beside full-height turnstile SITE EVIDENCE



Field-mounted access-control device SITE EVIDENCE



Protected biometric terminal - close view SITE EVIDENCE



Turnstile-side installation and conduit SITE EVIDENCE

Core Infrastructure & Support

BACK-END DELIVERY

Redundant servers, networking and power continuity

The application and database are hosted on dual servers with replicated data, supported by switching, rack infrastructure, backup power and controlled maintenance access.



Dual-server rack, switching and UPS SITE EVIDENCE



Secured production rack SITE EVIDENCE

1 YEAR

Comprehensive warranty and remote/on-site support

2 YEARS

CAMC after warranty, including quarterly preventive visits

24 / 48 HR

Complaint-resolution and priority breakdown response framework

5 YEARS

Essential spares support and long-term data backup

LIFECYCLE COVERAGE: 1-year warranty + 2-year CAMC = 3 years of structured service coverage

Customer Value & Proof

WHAT THIS HELPS THE CUSTOMER ACHIEVE

Safer gates. Faster administration. Better evidence.

The delivered platform turns workforce compliance into real-time access decisions and searchable management information.

01 Stronger site security

Live face verification reduces pass sharing and identity substitution.

02 Compliance at the gate

Expired or incomplete credentials stop access before entry.

03 Faster pass processing

Reusable masters, templates and controlled extensions reduce repeat work.

04 Accurate workforce visibility

Net-in, total-in, total-out and active-pass views support operations.

05 Defensible audits

Identity, documents, approvals, gate events and changes remain traceable.

06 Better contractor control

Labour-licence capacity and work-order mapping reduce uncontrolled passes.

07 Reliable attendance evidence

Entry-exit sequence supports man-hour and overtime analysis.

08 Scalable platform

Perpetual 5,000-user foundation supports future expansion.

PROJECT EVIDENCE

Order 4400000527 | Awarded 02 Dec 2022 | Approx. USD 0.048M

Project: Supply and Installation of Face Recognition System at Petronet LNG Limited, Dahej. USD value is a rounded presentation conversion of the awarded order value.

DELIVERED SUCCESSFULLY: SOFTWARE + HARDWARE + ACCESS + COMPLIANCE + SUPPORT

A Repeatable Industrial Solution

PETRONET LNG | DAHEJ LNG TERMINAL

Every worker verified.
Every pass governed.
Every movement traceable.

D-TECH combines customized software engineering, enterprise infrastructure and physical security integration to deliver one accountable gate-management platform.

REPLICABLE BLUEPRINT

Suitable for high-compliance industrial environments

The same architecture can be adapted for refineries, chemicals, pharmaceuticals, manufacturing, logistics parks, ports, utilities and large construction sites.

- LNG & ENERGY
- CHEMICALS
- PHARMA
- MANUFACTURING
- PORTS
- LOGISTICS

CUSTOMIZED SOFTWARE. SECURE ACCESS. AUDITABLE COMPLIANCE.